

TWO-DIMENSIONAL STRESS INTENSITY FACTOR COMPUTATIONS USING THE BOUNDARY ELEMENT METHOD

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SUMMARY

A multidomain boundary element formulation for the analysis of general two-dimensional plane strain/stress crack problems is presented. The numerical results were accurate and efficient. The analyses were performed using traction singular quarter-point boundary elements on each side of the crack tip(s) with and without transition elements. Traction singular quarter-point boundary elements contain the correct $\sqrt{r}$ displacement and $1/\sqrt{r}$ traction variations at the crack tip. Transition elements are appended to the traction singular elements to model the $\sqrt{r}$ displacement variation. The $1/\sqrt{r}$ traction singularity is not represented with these elements. Current research studies for the crack propagation analysis of quasi-static and fatigue fracture problems are discussed.

INTRODUCTION

The presence of cracks in a structure generally reduces the fatigue and static strength of the structure because the stresses and strains are highly magnified at the crack tip. It has been established that parameters deduced from linear elastic fracture mechanics can be used to determine the stress and strain magnification at the crack tip. These parameters, stress intensity factor (SIF), incorporate applied stress levels, geometry and crack size in a systematic manner and may be evaluated from the elastic stress analysis of cracked structures.

Two prominent numerical methods which are used to perform the elastic stress analysis are the finite element method (FEM) and the boundary element method (BEM). The BEM as used herein is sometimes called the 'boundary integral equation method' (BIEM). The FEM has been successfully used for elastic stress analysis and subsequent determination of the stress intensity factors (e.g. References 1 and 2). The BEM has also been successfully applied for computation of stress intensity factors (e.g. References 3-5).

The main disadvantage of the FEM is that a domain discretization is required to perform the analysis. The BEM involves discretization only of the boundary of the structure and the governing differential equation is solved exactly in the interior, leading to greater accuracy. However, the BEM generally leads to a nonsymmetric, fully populated system of equations, unlike the FEM which yields a symmetric, banded system of equations.

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A changing crack profile, as any problem which involves changing boundaries, requires that the discretization be redone at each step. The BEM involves discretizing the boundary only, whereas the FEM requires complete remeshing of the domain around the original and propagated crack. The BEM requires only the recomputation of the node co-ordinates along the crack (mode I and mixed-mode fracture problems) and the addition of one or two elements (mixed-mode fracture problems) to represent an extended crack profile.

Several strategies have been proposed for the analysis of crack problems using the boundary element method. These methods include representing the crack as a notch, symmetric crack modelling, use of special Green's functions, and flat crack modelling. Representing the crack as a notch or replacing the crack plane with symmetry boundary conditions, i.e. symmetric crack modelling, removes the singularity in the algebraic system of equations which is obtained when the upper and lower crack surfaces are modelled in the same plane.⁶ It is, however, limited to symmetric crack problems. Representing the crack as a notch increases the modelling error due to the notch opening. The special Green's function approach^{3-5,7} possesses the advantages that crack geometry and crack tip singularities are fully embedded; no modelling of the crack surface is required. The disadvantage is that some two-dimensional and all three-dimensional problems cannot be formulated using the special Green's function approach.⁴ Also, complex arithmetic is required for the problems for which it does apply. Flat crack modelling⁴ represents the displacements along the crack surface as the relative displacement between the two crack surfaces. This scheme has two critical deficiencies as a mathematical model for crack geometries, as pointed out by Cruse.⁴ First, if there are no tractions on the outer boundary and only crack surface loading, a non-unique boundary integral equation is generated. Second, two unknown displacement variables, i.e. the relative and total displacements, exist along the crack. A possible solution for these problems is given by Cruse.⁴ An extensive discussion of the special Green's function and flat crack modelling techniques is given by Cruse.⁴

This paper presents alternative strategies for modelling the crack and subsequent SIF computations. The essence of the present approach is a multidomain discretization using isoparametric linear and/or quadratic boundary elements. Multidomain discretization eliminates the two problems of flat crack modelling, i.e. non-uniqueness of the boundary integral equation resulting from crack surface loading and the problem of two displacement variables along the crack, because each crack surface is modelled in a separate domain. Therefore, the prescribed tractions and the unknown displacements along the crack surfaces are independently represented in each domain. Furthermore, the crack representation is mathematically correct, i.e. the crack is not represented as a notch.

The isoparametric quadratic boundary elements are transformed to properly represent the analytically dominant $\sqrt{r}$ and $1/\sqrt{r}$ behaviour exhibited by the displacements and tractions, respectively, in the vicinity of the crack tip. The transformations used in this paper are performed in real space which eliminates the complex arithmetic required in the special Green's function approach. Furthermore, the concepts developed in this paper are valid for all two-dimensional linear elastic fracture problems and the techniques can be extended to three-dimensional problems.

The stress intensity factors are computed automatically within the program after performing the elastic analysis of the structure. The stress intensity factors are computed using the displacement correlation technique of References 8-10. This technique fully exploits the traction singular quarter-point boundary element interpolations.

Necessary aspects of linear elastic fracture mechanics are presented in the first section of the paper. Then the multidomain boundary integral equation for two-dimensional plane strain/stress is presented. The displacements and tractions are then represented using iso-

parametric linear and/or quadratic boundary elements yielding the boundary element formulation of the problem. The details of the quadratic element transformation to traction singular quarter-point and transition boundary elements are presented. The stress intensity factor computational scheme is given. Example results are provided to verify the validity of the formulation.

LINEAR ELASTIC FRACTURE MECHANICS

Linear elastic fracture mechanics analysis neglects the inelasticity at the crack tip, such as microcracking and plasticity. The inelastic effects are considered to be restricted to a sufficiently small volume such that they can be neglected. Linear elastic fracture mechanics relates the stress field magnitude and distributions in the vicinity of the crack tip to the applied nominal stress and to the geometry of the cracked structure.

Irwin¹¹ observed that there are three independent kinematic movements of the upper and lower crack surfaces with respect to each other: (1) the crack opening mode (mode I); (2) the crack sliding mode (mode II) and (3) the crack tearing mode (mode III). This paper is limited to two-dimensional plane strain/stress analysis; therefore, only modes I and II are considered.

Considering a crack problem in an infinite domain, Williams¹² has shown that the stresses for the traction-free crack may be written in terms of an infinite series with respect to the polar co-ordinates r and θ . Substituting Irwin's¹¹ definition of the stress intensity factors into

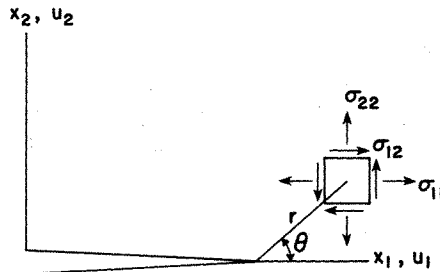


Figure 1. Crack in an infinite plane

Williams'¹² eigenfunction expansion yields (see Figure 1):

$$\begin{aligned}\sigma_{11} &= \frac{K_I}{\sqrt{2\pi r}} \cos \frac{\theta}{2} \left(1 - \sin \frac{\theta}{2} \sin \frac{3\theta}{2} \right) - \frac{K_{II}}{\sqrt{2\pi r}} \sin \frac{\theta}{2} \left(2 + \cos \frac{\theta}{2} \cos \frac{\theta}{2} \right) + \dots \\ \sigma_{22} &= \frac{K_I}{\sqrt{2\pi r}} \cos \frac{\theta}{2} \left(1 + \sin \frac{\theta}{2} \sin \frac{3\theta}{2} \right) + \frac{K_{II}}{\sqrt{2\pi r}} \sin \frac{\theta}{2} \cos \frac{\theta}{2} \cos \frac{3\theta}{2} + \dots \\ \sigma_{12} &= \frac{K_I}{\sqrt{2\pi r}} \cos \frac{\theta}{2} \sin \frac{\theta}{2} \cos \frac{3\theta}{2} + \frac{K_{II}}{\sqrt{2\pi r}} \cos \frac{\theta}{2} \left(1 - \sin \frac{\theta}{2} \sin \frac{3\theta}{2} \right) + \dots\end{aligned}\quad (1)$$

in which K_I and K_{II} are the mode I and mode II SIF's, respectively, and r, θ are defined in Figure 1 along with the stresses.

Integrating equations (1) using the strain–displacement and stress–strain relations yields the following relationships for displacements relative to the crack tip for $r \ll a$.^{11,13}

$$\begin{aligned} u_1 &= \frac{K_I}{4\mu} \sqrt{\frac{r}{2\pi}} \left[(2\kappa - 1) \cos \frac{\theta}{2} - \cos \frac{3\theta}{2} \right] + \frac{K_{II}}{4\mu} \sqrt{\frac{r}{2\pi}} \left[(2\kappa + 3) \sin \frac{\theta}{2} + \sin \frac{3\theta}{2} \right] \\ u_2 &= \frac{K_I}{4\mu} \sqrt{\frac{r}{2\pi}} \left[(2\kappa - 1) \sin \frac{\theta}{2} - \sin \frac{3\theta}{2} \right] - \frac{K_{II}}{4\mu} \sqrt{\frac{r}{2\pi}} \left[(2\kappa - 3) \cos \frac{\theta}{2} + \cos \frac{3\theta}{2} \right] \end{aligned} \quad (2)$$

in which μ = shear modulus = $E/[2(1+\nu)]$; E = Young's modulus; ν = Poisson's ratio; $\kappa = (3-4\nu)$ for plane strain; and $\kappa = (3-\nu)/(1+\nu)$ for plane stress.

For accurate modelling of stresses and displacements, any numerical approximation must take into account the $\sqrt{r}$ and $1/\sqrt{r}$ displacement and stress variations, respectively. It will be shown that the boundary elements adjacent to crack tip do possess the correct displacement and stress variations.

BOUNDARY INTEGRAL EQUATION

The boundary integral equation formulation discretizes only the boundary and thus effectively reduces the dimension of the calculation by one. The governing Navier–Cauchy equations of elasticity are replaced with a boundary integral operator using Betti's reciprocal theorem.^{14,15}

Dividing the plane into subdomains $\Omega^{(k)}$, the boundary integral equation is written as:^{16†}

$$C_{ij}^{(k)}(P)u_j^{(k)}(P) + \int_{\Gamma^{(k)}} T_{ij}^{(k)}(P, Q)u_j^{(k)}(Q) d\Gamma = \int_{\Gamma^{(k)}} U_{ij}^{(k)}(P, Q)t_j^{(k)}(Q) d\Gamma \quad (3)$$

in which $C_{ij}^{(k)}(P)$ is the rigid body translation solution of equation (3).^{17,18} For a smooth boundary, $C_{ij}^{(k)}(P) = \frac{1}{2}\delta_{ij}$, in which δ_{ij} is the Kronecker delta. The points P and Q are field points on the boundary, $\Gamma^{(k)}$. The second-order tensors $T_{ij}^{(k)}(P, Q)$ and $U_{ij}^{(k)}(P, Q)$ are the fundamental traction and displacement solutions, respectively, which are:^{14,15}

$$\begin{aligned} T_{ij}^{(k)}(P, Q) &= -\frac{1}{4\pi\mu^{(k)}(\tau^{(k)} + \mu^{(k)})} \left(\frac{1}{r(P, Q)} \right) \\ &\times \left\{ \frac{\partial r}{\partial n^{(k)}(Q)} (2(\mu^{(k)})^2)\delta_{ij} + 4\tau^{(k)}\mu^{(k)}r_{,i}(P, Q)r_{,j}(P, Q) \right. \\ &\left. + 2(\mu^{(k)})^2[r_{,j}(P, Q)n_i^{(k)}(Q) - r_{,i}(P, Q)n_j^{(k)}(Q)] \right\} \end{aligned} \quad (4)$$

$$U_{ij}^{(k)}(P, Q) = -\frac{1}{4\pi\mu^{(k)}(\tau^{(k)} + \mu^{(k)})} \{ (\tau^{(k)} + 2\mu^{(k)}) \ln r(P, Q)\delta_{ij} - \tau^{(k)}r_{,i}(P, Q)r_{,j}(P, Q) \}$$

in which $\tau^{(k)} = \mu^{(k)}/(1-2\nu^{(k)})$ for plane strain and $\tau^{(k)} = 2\mu^{(k)}(1+\nu^{(k)})$ for plane stress. The distance measured from point P to point Q is $r(P, Q)$, and $n^{(k)}$ is the unit outward normal to the boundary at point Q . The tractions, $t_i^{(k)}$, are related to the stresses as:

$$t_i^{(k)} = \sigma_{ij}^{(k)} n_j^{(k)}$$

The fundamental solutions given in equations (4) represent the traction and displacement in the j th direction at point Q due to a unit point force in the i th direction at point P .

† The usual indicial notation of Cartesian tensor analysis is used. Latin subscripts have the range (1, 2) and summation over repeated subscripts is implied. Commas are used to denote partial differentiation.

At a point along the interface between subdomains $\Omega^{(k)}$ and $\Omega^{(l)}$, the displacement continuity and the traction equilibrium are enforced through

$$\begin{aligned} u_i^{(k)}(P) &= u_i^{(l)}(P) \\ t_i^{(k)}(P) &= -t_i^{(l)}(P) \end{aligned} \quad (5)$$

MULTIDOMAIN BOUNDARY ELEMENT METHOD

The isoparametric BEM represents the geometric, displacement and traction variations as polynomials:

$$\begin{aligned} x_i &= \sum_{\beta=1}^{n_\beta} N^\beta(\xi) x_i^\beta = [N(\xi)] \{x_i\} \\ u_i &= [N(\xi)] \{u_i\} \\ t_i &= [N(\xi)] \{t_i\} \end{aligned} \quad (6)$$

in which $n_\beta = 2$ for the linear boundary elements and $n_\beta = 3$ for the quadratic boundary elements, $[N(\xi)]$ is the row vector of shape functions in terms of the nondimensional co-ordinate ξ as defined in Figure 2, and the terms in braces are the element nodal values of the parameter.

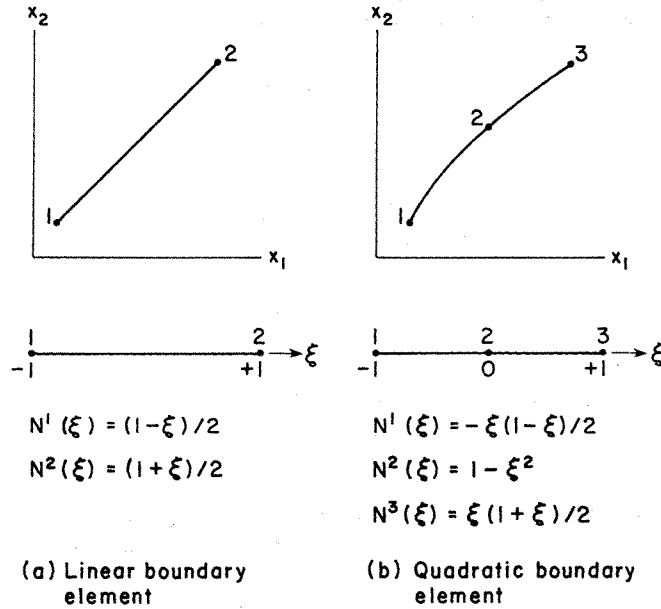


Figure 2. Boundary elements

Substituting the polynomial expressions of equations (6) into equation (3) results in the multidomain boundary element formulation:

$$\begin{aligned} C_{ij}^{(k)}(P_\eta) u_j^{(k)}(P_\eta) + \sum_{\sigma=1}^{M(k)} \left[\int_{-1}^1 T_{ij}^{(k)}(P_\eta, Q(\xi)) N^\beta(\xi) J_\Gamma(\xi) d\xi \right] (u_j^{(k)})^{\beta\sigma} \\ = \sum_{\sigma=1}^{M(k)} \left[\int_{-1}^1 U_{ij}^{(k)}(P(\eta), Q(\xi)) N^\beta(\xi) J_\Gamma(\xi) d\xi \right] (t_j^{(k)})^{\beta\sigma} \end{aligned} \quad (7)$$

in which P_n is the fixed boundary point, $Q(\xi)$ is the variable boundary point, $M^{(k)}$ is the number of elements in subdomain k , $J_T(\xi)$ is the boundary Jacobian, i.e.

$$J_T(\xi) = \sqrt{\left[\left(\frac{dx_1}{d\xi}\right)^2 + \left(\frac{dx_2}{d\xi}\right)^2\right]} = \sqrt{\left[\left(\frac{dN^\beta}{d\xi}x_1^\beta\right)^2 + \left(\frac{dN^\beta}{d\xi}x_2^\beta\right)^2\right]}$$

and $(u_j^{(k)})^{\beta\sigma}$, $(t_j^{(k)})^{\beta\sigma}$ are the element node components of displacement and traction, respectively. Evaluating equation (7) for each boundary node for each subdomain and imposing the interface constraint conditions of equations (5), leads to an algebraic system of equations.

CRACK ELEMENTS FOR THE BEM

Using the elements shown in Figure 2 in the vicinity of the crack tip gives either a linear or quadratic (depending on which element is used) variation of the displacements and tractions. The crack tip displacements and tractions are of order $\sqrt{r}$ and $1/\sqrt{r}$, respectively, as shown in equations (1) and (2). Since the interpolating polynomials do not contain these terms, the tractions can never be represented correctly by the interpolating functions and the displacements can be adequately modelled only by using a refined mesh in the vicinity of the crack tip.

It has been shown in the finite element literature^{1,2,19} that the $\sqrt{r}$ displacement variation can be represented using isoparametric quadratic elements by placing the midpoint node at the quarter-point, as shown in Figure 3. Using the shape function definitions given in Figure 3(b), the variation of the displacements and tractions then become:

$$\begin{matrix} u_i \\ t_i \end{matrix} = A_i^1 + A_i^2\sqrt{r} + A_i^3r \quad (8)$$

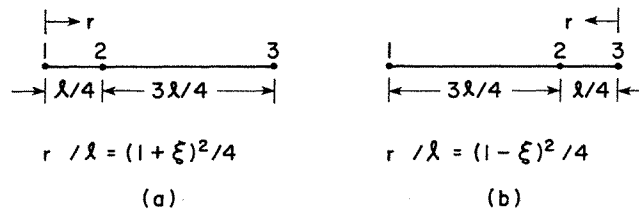


Figure 3. Traction singular quarter-point boundary elements

In finite element applications, the tractions are obtained by differentiating the displacement interpolation functions, which results in the correct $1/\sqrt{r}$ singularity in the traction variations being obtained. However, in the BEM the displacement and traction variations are independently represented.

The displacement variation given by equation (8) is analytically correct for the first terms of the infinite series expansion, equation (2). However, the traction variation does not possess the correct $1/\sqrt{r}$ singularity. The inclusion of this singularity is accomplished by multiplying the right-hand side of equation (8) by $\sqrt{(l/r)}$:²⁰

$$t_i = (A_i^1 + A_i^2\sqrt{r} + A_i^3r)\sqrt{(l/r)} = B_i^1/\sqrt{r} + B_i^2 + B_i^3\sqrt{r} \quad (9)$$

which corresponds to the correct traction variations in the vicinity of the crack tip, equation (1). The expression $\sqrt{(l/r)}$ in equation (9) is written in the nondimensionalized co-ordinate system, as shown in Figures 3(a) and 3(b),

Thus, the displacements and traction variations for the elements adjoining the crack tip are given by equations (8) and (9). The resulting boundary elements are 'traction singular quarter-point' boundary elements.

The elements adjoining the traction singular quarter-point boundary elements are the standard boundary elements shown in Figure 2. Improved results with coarse meshes may be obtained using transition elements.¹⁹ The function of the transition elements is to extend the distance over which the crack is sensed.

Consider the element shown in Figure 4. The three nodes of this element, designated by the nondimensional variable $s = 0, p, 2$, have been mapped on to the ξ -axis. Lynn and Ingraffea¹⁹ have shown that by placing the midpoint node at

$$p = \frac{(1-q) + \sqrt{(q^2 + 2q)}}{2}$$

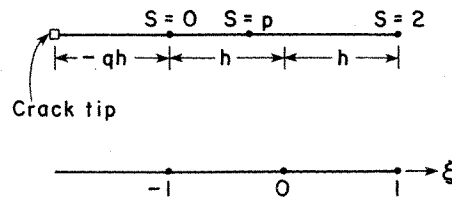


Figure 4. Transition boundary element co-ordinate mapping

the correct stress singularity is obtained by appropriate differentiation of the element shape functions. For the boundary element analysis, the following displacement and traction variations are obtained:

$$\begin{matrix} u_i \\ t_i \end{matrix} = C_i^1 + C_i^2 \sqrt{r} + C_i^3 r \quad (10)$$

in which r is measured from the crack tip, at point $-qh$ in Figure 4. Notice that equation (10) has the same form as equation (8). The technique used to transform the quarter-point boundary elements into traction singular quarter-point boundary elements, i.e. multiplication by $\sqrt{(l/r)}$, is not applicable for transition elements because the continuity of the traction modelling would be destroyed. Therefore, the analyses which use transition elements correspond to using the interpolation functions given by equation (10). The use of such transition elements correctly represents the displacement functions but does not adequately model the tractions. Thus, transition element representations must be tested completely in order to determine their usefulness.

COMPUTATION OF STRESS INTENSITY FACTORS

The displacement correlation method is employed for the computation of the stress intensity factors. The displacement correlation method used herein is that given by References 8–10. The method fully exploits the traction singular quarter-point boundary element interpolations. Denoting the displacements along the crack axis as u (crack sliding displacement, CSD) and the displacements normal to the crack axis as v (crack opening displacement, COD) the variations of

the CSD and COD are, using equation (8),

$$\begin{aligned} u &= u_A + (-3u_A + 4u_B - u_C)\sqrt{(r/l)} + (2u_A - 4u_B + 2u_C)r/l \\ v &= v_A + (-3v_A + 4v_B - v_C)\sqrt{(r/l)} + (2v_A - 4v_B + 2v_C)r/l \end{aligned} \quad (11)$$

in which the points A, B and C are defined in Figure 5(a). The expressions in equations (11) define the generalized constants A_i^1 , A_i^2 , A_i^3 in equation (8). For a symmetric crack problem, $u = v_A = 0$ and equation (11) reduces to

$$v = (4v_B - v_C)\sqrt{(r/l)} + (-4v_B + 2v_C)r/l \quad (12)$$

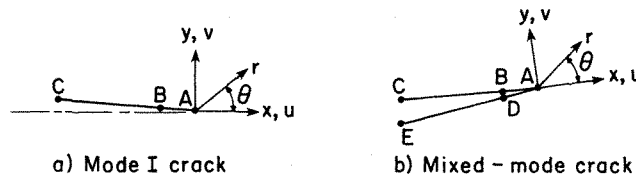


Figure 5. Element geometries for SIF computations

The coefficient multiplying $\sqrt{r}$ in equation (12) is equated to the u_2 expression given in equation (2). Evaluating equation (2) for $\theta = 180^\circ$ gives:^{8,9}

$$K_I = \frac{2\mu}{\kappa + 1} \sqrt{\frac{2\pi}{l}} (4v_B - v_C) \quad (13)$$

The general case shown in Figure 5(b) for $\theta = 180^\circ$ decouples the interaction of K_I and K_{II} , see equation (2), to give:¹⁰

$$\begin{aligned} K_I &= \frac{\mu}{\kappa + 1} \sqrt{\frac{2\pi}{l}} [4(v_B - v_D) + v_E - v_C] \\ K_{II} &= \frac{\mu}{\kappa + 1} \sqrt{\frac{2\pi}{l}} [4(u_B - u_D) + u_E - u_C] \end{aligned} \quad (14)$$

in which the points B, C, D and E are defined in Figure 5(b). Equations (14) are used to evaluate the mixed-mode stress intensity factors.

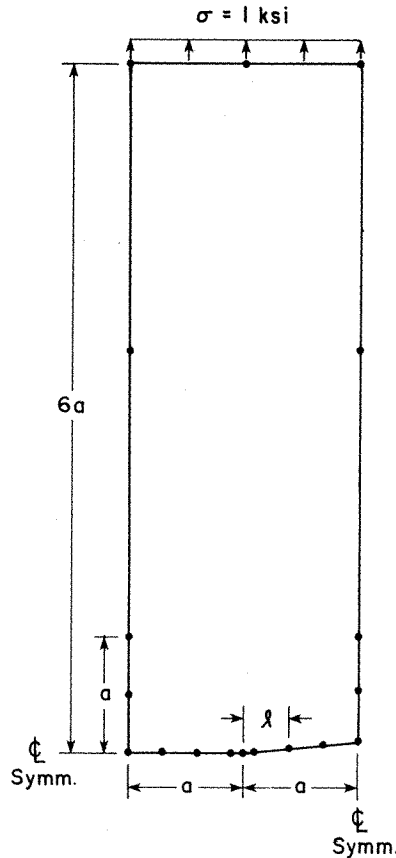
EXAMPLE RESULTS

The material values for the problems in this section were taken to be $E = 5250$ ksi and $\nu = 0.20$, which are typical of the values found in rock.

The examples are either mode I or mixed-mode, and cover a range of problems in linear elastic fracture mechanics. Both tension and bending loaded mode I problems are considered because of the differences in the structure behaviour. The numerical solutions are compared with collocation, finite element and other boundary element solutions. All problems were analysed using double precision on the IBM 370/168 computer at Cornell University.

Double edge cracked plate (Figure 6)

The double edged cracked plate problem was analysed by Bowie²¹ using a collocation procedure. This problem is doubly symmetric and only one-quarter of the plate needs to be

Figure 6. Double edged crack problem ($a = 1.80$ in.)

discretized, as shown in Figure 6. The plate is subjected to a uniform tension of magnitude σ .

The problem was analysed using (i) quadratic, (ii) traction singular quadratic, (iii) quarter-point and (iv) traction singular quarter-point boundary elements for the two elements adjoining the crack tip and quadratic elements elsewhere. Traction singular quadratic elements are boundary elements which include the $\sqrt{(l/r)}$ modification in the traction variations without moving the midpoint node to the quarter-point position. Quarter-point elements are boundary elements in which the central node is placed at the quarter-point position only, i.e. no traction interpolation modification.

The results obtained for various crack tip element to crack length ratios (l/a) and a fixed number of degrees-of-freedom are presented in Figure 7. These results are compared with Bowie's²¹ collocation result (accurate within ± 1 per cent):

$$K_I = C_L C_W \sigma \sqrt{(\pi a)} = (1.145)(1.128)(2.378) = 3.071$$

in which C_L is the finite length correction factor, C_W is the finite width correction factor and $\sigma \sqrt{(\pi a)}$ is the exact solution for an infinite plate. The numbers for C_L and C_W are given for a length to width ratio of $L/W = 3$.

The results obtained using the four types of boundary elements are compared with the collocation solution in Figure 7. The quadratic and traction singular element results were

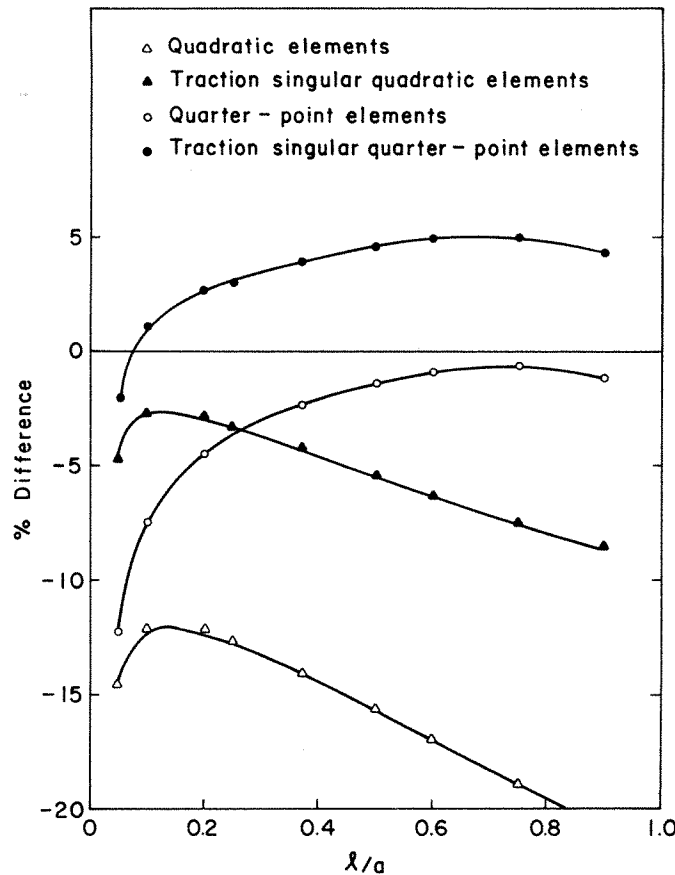


Figure 7. Percentage difference in the calculated K_I for the double edge crack problem

obtained using displacement correlations at the midpoint node; the quarter-point and traction singular quarter-point element results were correlated using equation (13).

The plotted results in Figure 7 reveal that the best accuracy obtained using quadratic elements is 12 per cent for an $l/a = 0.10$. The traction singular element results give a percentage difference of 3 per cent for $l/a = 0.10$. The quadratic and traction singular elements yield increasingly inaccurate results for $l/a > 0.10$. The quarter-point element results show convergence from an initial difference of 12.7 per cent for $l/a = 0.05$ to a difference of only 0.60 per cent for $l/a = 0.75$ and then diverges slightly. The traction singular quarter-point elements results are quite accurate (less than 3 per cent) for $0.05 \leq l/a \leq 0.25$, with a maximum difference of 5 per cent at $l/a = 0.75$.

These results show that traction singular quarter-point elements should be used within the range of $0.05 \leq l/a \leq 0.25$. The other type of boundary elements should not be used since they result in inferior results in this range and do not properly represent both the $\sqrt{r}$ displacement behaviour and $1/\sqrt{r}$ traction behaviour. It could be argued that the quarter-point boundary elements should be used with large l/a ratios, but such large l/a ratios invalidate the assumption of small r used in truncating the infinite series expansion for the stresses and displacements to only one term, equations (1) and (2),

Centre cracked plate (Figure 8)

The square, centre cracked plate problem of Figure 8(a) was analysed using an increasing number of quadratic elements for a fixed $l/a = 0.10$. The results are plotted in Figure 8(b) using traction singular quarter-point elements at the crack tip (the two elements contained within the circled nodes) with and without transition elements. All elements, excluding the two crack tip elements, are uniformly decreased in length from the basic mesh shown in Figure 8(a), which corresponds to the six element case plotted in Figure 8(b).

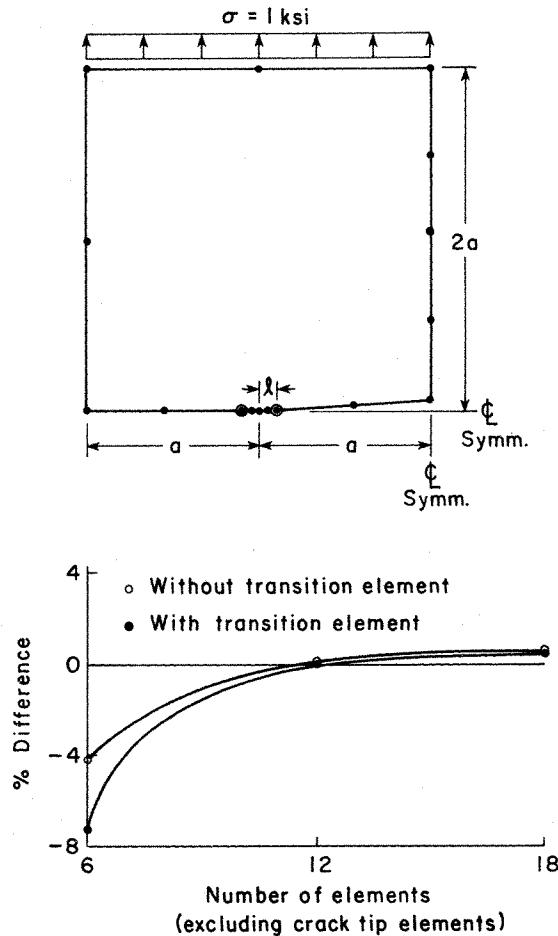


Figure 8. Centre crack square plate problem: (a) Typical boundary element mesh ($a = 1.0$ in.); (b) Convergence curves

The plotted results are compared with the collocation results of Isida²² which are accurate within ± 1 per cent. Figure 8(b) shows that the results obtained using transition elements are slightly better than the results obtained without transition elements. This problem further shows that very accurate results are obtained using only a few boundary elements.

Four point bend specimen (Figure 9)

The four point beam bending problem is singly symmetric about the crack axis. The boundary element discretization is shown in Figure 9.

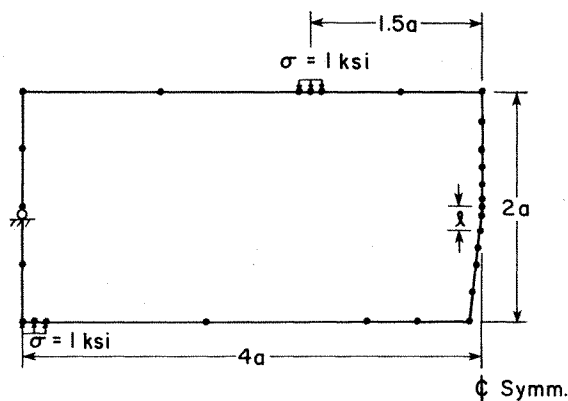


Figure 9. Four point bend crack problem ($a = 1.0$ in.)

The moment loaded crack problem, Figure 9, was analysed using a fixed number of degrees-of-freedom and various l/a ratios. The two end elements along the crack were fixed in length to be $0.50a$. The results of the analysis are given in Table I. BEM-1 represents the solutions obtained without transition elements and BEM-2 represents the solutions obtained with transition elements. These results are compared with the results of Brown and Srawley:²³

$$K_I = \frac{6Ma^{1/2}}{BW^2} [1.99 - 2.47(a/W) + 12.97(a/W)^2 - 23.17(a/W)^3 + 24.80(a/W)^4]$$

in which M is the applied bending moment, W is the plate width and B is the plate thickness. The formula given by Brown and Srawley is accurate to within ± 1 per cent for the range $0 < a/W \leq 0.60$. The results given in Table I reveal that the present boundary element method yields accurate stress intensity factors for the range $0.10 \leq l/a \leq 0.20$, as contrasted with FEM results² for bending loaded crack problems in which very small l/a ratios were used.

Compact specimen (Figure 10)

The boundary element mesh for the single symmetric compact specimen is shown in Figure 10. The compact specimen was analysed for a range of $\alpha = a/W$ values using a fixed number of

Table I. BEM solutions for the four point bend crack problem

l/a	BEM-1	% Diff.	BEM-2	% Diff.
0.05	3.883	1.70	3.949	3.43
0.10	3.870	1.36	3.893	1.96
0.15	3.816	-0.05	3.834	0.04
0.20	3.759	-1.55	3.777	-1.07
0.25	3.705	-2.96	3.725	-2.44
Brown and Srawley ²³			3.818	

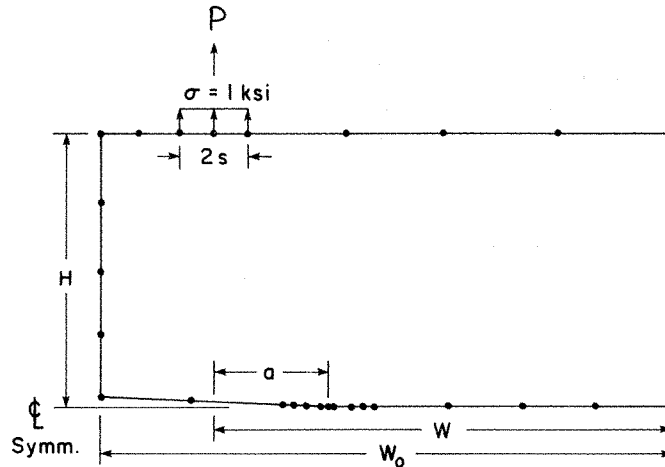


Figure 10. Compact specimen problem ($W_0 = 1.25W$, $H = 0.60W$, $2s = 0.125W$, $W = 4.00$ in.)

degrees-of-freedom. This problem was analysed to test the boundary element formulation of this paper on a high degree of bending problem for a wide range of crack lengths.

Two sets of boundary element results were generated: (1) analyses for a fixed crack tip element length of $l = 0.10$ in. and (2) analyses for $l/a = 0.05$. The boundary element results are compared with the results of Srawley.²⁴

$$K_I = \frac{P}{B\sqrt{W}} \frac{(2 + \alpha)(0.886 + 4.64\alpha - 13.32\alpha^2 + 14.72\alpha^3 - 5.64\alpha^4)}{(1 - \alpha)^{3/2}}$$

in which P is the applied force. The formula given by Srawley is accurate within ± 1 per cent for the range $0.20 \leq \alpha \leq 1.00$. The Srawley formula is an interpolation of the 'analytical results considered to be most accurate and reliable'.²⁴ The results given in Figure 11 show that the BEM is accurate (percentage difference ≤ 2.5 per cent) for the range $0.30 \leq \alpha \leq 0.70$ using either of the discretizations with the results corresponding to $l = 0.10$ in. being more accurate. The two extreme cases shown in Figure 11, i.e. $\alpha = 0.20$ and $\alpha = 0.80$, are accurate for the discretization which gives the smaller crack tip element length. For the case $\alpha = 0.20$, the results appear to be sensitive due to the proximity of the applied loading, and for $\alpha = 0.80$ the results appear to be sensitive due to the proximity of the external boundary.

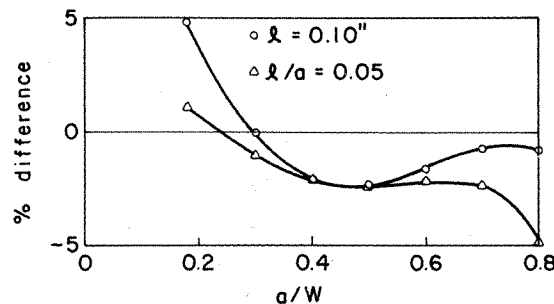


Figure 11. Percentage difference in calculated K_I for the compact specimen

Centre slant crack

Two mixed-mode centre slant crack problems were analysed. The first, shown in Figure 12, was analysed with an increasing number of nonsingular quadratic elements to determine the convergence characteristics for the mixed-mode problems. The numerical results were obtained

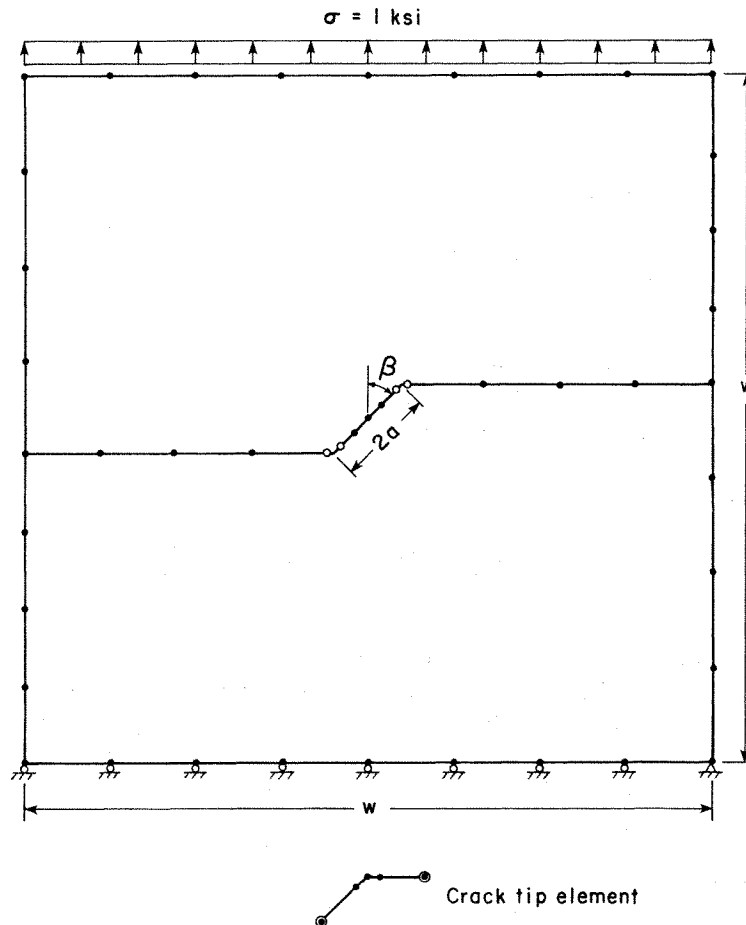


Figure 12. Typical boundary element mesh for the 'infinite' centre slant crack problem ($w = 10$ in., $2a = \sqrt{2}$ in., $\beta = 45^\circ$)

for a fixed $l/a = 0.10$ at the two crack tips. This problem is representative of a crack in an infinite domain. The results are compared with the exact results of a crack in an infinite plane:

$$K_I = \sigma \sqrt{(\pi a)} \sin^2 \beta$$

$$K_{II} = \sigma \sqrt{(\pi a)} \cos \beta \sin \beta$$

in which β is the angle between the vertical load axis and the crack axis. Since this crack problem is not truly infinite, finite length and width correction factors should be applied to the infinite plate results. However, these correction factors would be quite small on the basis of results obtained for mode I problems (see, for example, Reference 25). Figure 13 shows that the

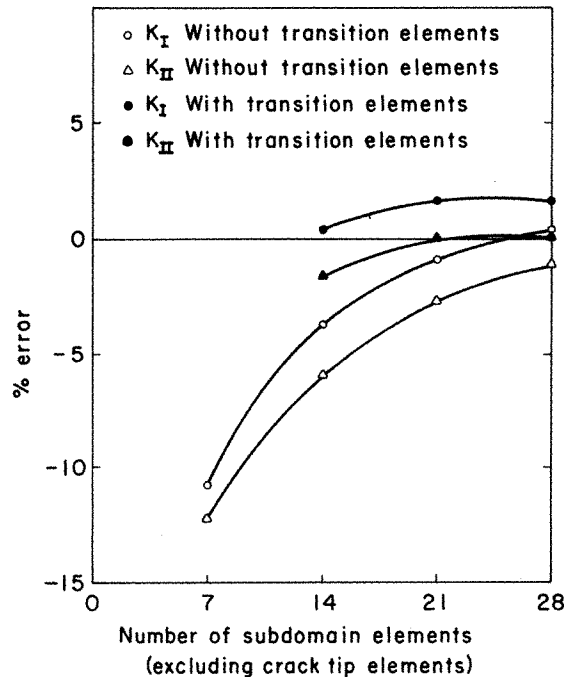


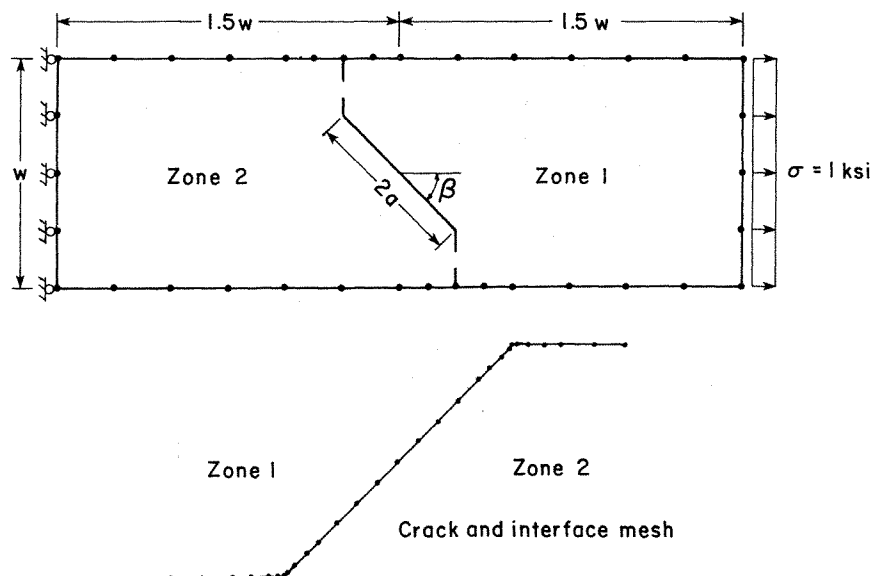
Figure 13. Percent error in the computed SIF for the 'infinite' centre slant crack problem

convergence curve without transition elements has a much steeper slope than the results with transition elements. Yet the difference between K_I and K_{II} convergence curves using transition elements is about the same as the difference between the K_I and K_{II} convergence curves without transition elements. The transition element results are good for the last two boundary element meshes. These results show that better results are obtained using a coarse mesh with transition elements than without transition elements.

The second mixed-mode centre slant crack example, Figure 14, is a finite width and length plate problem. The problem was originally analysed by Tada *et al.*²⁵ using the collocation method. The boundary element mesh is shown in Figure 14 and the dashed lines are used to represent the interface. The interface constraint equations, equations (5), are applied along these lines but not along the crack. The normalized stress intensity factors are given in Table II for the special Green's function approach of Cruse and Wilson,²⁰ the finite element results reported by Cruse and Wilson²⁰ using the CHILES finite element program and the results of the writers' boundary element formulation without transition elements. Table II shows that very accurate results are obtained using the present boundary element formulation. The CPU times are qualitative since the writers do not know which computer Cruse and Wilson used.

SUMMARY AND CONCLUSIONS

The use of multidomain boundary element analysis allows the crack surfaces to be discretized. Since the crack surfaces are independently represented in each domain, they can displace independently of one another and the tractions can be independently specified. This avoids the matrix singularity problems discussed by Cruse⁶ and the non-uniqueness of flat crack modelling.⁴

Figure 14. Centre slant crack problem ($w = 1.0$ in., $2a = \sqrt{2}$ in., $\beta = 45^\circ$)

The purpose for computing stress intensity factors is the determination of fracture initiation and subsequent propagation. Work is currently being performed by the writers for the automatic propagation of cracks using the boundary element method. The boundary element method is used to perform both quasi-static and fatigue crack propagation analysis for mode I and mixed-mode fracture problems. The results of this work will be reported in later papers.

The use of traction singular quarter-point boundary elements with l/a ratios between 0.05 and 0.20 was shown to yield accurate stress intensity factors for both mode I and mixed-mode problems. This range of l/a ratios was also shown to be accurate for bending loaded crack problems.

The wide range stress intensity factor results for the compact specimen were shown to differ by less than 2.5 per cent when compared with the interpolated results of Srawley²⁴ for a range of $0.30 \leq a/W \leq 0.70$. The cases corresponding to $a/W = 0.20$ and $a/W = 0.80$ showed larger discrepancies in the computed results, with the smaller crack tip element length results giving the

Table II. Centre slant crack problem results

SIF†	Ref. 25	Ref. 20	Ref. 20‡	BEM
F_I	0.730	0.712	0.728	0.725
F_{II}	0.600	0.590	0.590	0.598
% Diff.	—	-2.47	-0.27	-0.68
	—	-1.67	-1.67	-0.33
CPU (sec)	—	46	3+48	19

† Normalized by dividing by $\sigma\sqrt{\pi a}$.

‡ Using the CHILES finite element program.

better solutions. The larger crack tip element length results for the two extreme cases, i.e. $a/W = 0.20$ and $a/W = 0.80$, differed by less than 5 per cent when compared with the interpolated results of Srawley.²⁴

The use of transition elements in conjunction with the traction singular quarter-point elements gives improved results with coarse meshes for the mixed-mode crack problems. Further, it was shown that the convergence curves for the mixed-mode stress intensity factors with transition elements are flatter than the convergence curves for the mixed-mode stress intensity factors without transition elements. This means that the computed mixed-mode stress intensity factors which include transition boundary elements are not as discretization sensitive as are the stress intensity factor computations based on the elastic analysis without transition elements. The use of transition elements for mode I problems yield approximately the same results as are obtained without using transition elements for both tension and bending loaded crack problems.

The numerical results cited in this paper illustrate the scope of fracture mechanics analysis problems that can be modelled using the BEM. The accuracy of the results is clearly sufficient for fracture initiation predictions, particularly for mixed-mode fracture initiation predictions. Even greater accuracy is obtainable with the BEM by using higher order (higher than quadratic) polynomial interpolation in the crack tip vicinity. The transformation of the higher order interpolation functions to contain the dominant displacement variation is given in Hibbitt.²⁶ The incorporation of the dominant traction singularity would be performed using the same techniques outlined in this paper.

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